

Relationship between Fröhlich and Davydov models of biological order

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Two physical models to explain the unusual behavior of biological systems put forward by Fröhlich and Davydov are reviewed. The Davydov Hamiltonian is transformed into normal coordinates and the Fröhlich Hamiltonian is canonically transformed into an equivalent form within the Hartree-Fock approximation. A model Hamiltonian capable of relating both theories is proposed. The resulting analysis employs only one type of second-quantized operators which describe the effect on the active dipolar modes of vibration in biological cells. An equation of motion is derived for the creation and annihilation operators that are used. When the phenomenon of Bose condensation is taken into account, the field translation variable is shown to obey a nonlinear Schrödinger equation. Its solutions are provided and their significance is discussed. Some of the solutions are shown to correspond to properties already examined by Fröhlich and Davydov. New solutions are found and interpreted as interfaces between either an ordered and a disordered phase or two ordered phases existing in the biosystem. The relationship between the two theories of biological order is demonstrated and discussed.

I. THE FRÖHLICH MODEL

Fröhlich presented, in a series of papers,¹⁻⁷ a quantum theory of long-range coherence in biological cells predicting the existence of coherent elastic vibrations in the microwave region (10^{11} – 10^{12} Hz). The effect presupposes

- (i) the existence of a large transmembrane potential difference,
- (ii) a continuous supply of metabolic energy within a rate of supply exceeding a critical value, and
- (iii) internal organization of the biosystem resulting from a spatial arrangement and a sufficient strength for the coupling constants between the component parts.

The biosystem is schematically divided into three interacting subsystems (Fig. 1):

- (i) an oscillating segment of giant dipoles located along the chain formed by macromolecules,
- (ii) a heat bath (containing all water and suspended ions), and
- (iii) an external source of energy supply.

Following Fröhlich,⁷ Wu and Austin⁸⁻¹¹ introduced the following Hamiltonian to adequately describe the above situation:

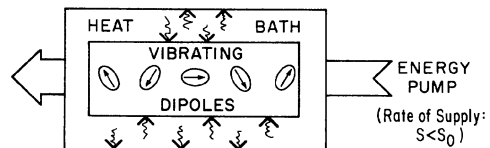
$$\begin{aligned}
 H = & \sum_i \omega_i a_i^\dagger a_i + \sum_i \Omega_i b_i^\dagger b_i + \sum_i \theta_i P_i^\dagger P_i \\
 & + \frac{1}{2} \sum_{i,j,k} (\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k) \\
 & + \sum_{i,j} (\lambda b_i a_j^\dagger + \lambda^* b_i^\dagger a_j) + \sum_{i,j} (\xi P_i a_j^\dagger + \xi^* P_i^\dagger a_j), \quad (1)
 \end{aligned}$$

where $(a_i^\dagger, a_i)$, $(b_i^\dagger, b_i)$, and $(P_i^\dagger, P_i)$ are, respectively, the cell, heat-bath, and energy-pump creation and annihilation (boson-type) operators.

Provided that the oscillating dipoles are within a narrow band of resonance frequencies ($\omega_{\min} \leq \omega_i \leq \omega_{\max}$) and that the coupling constants χ , λ , and ξ are large enough, the following effects are expected to occur.

- (i) The existence of a strong, long-range ($\sim 1 \mu\text{m}$) attractive forces between the dipoles. The effective Fröhlich potential between two interacting dipoles which vibrate with frequencies $\omega_1 = \omega_2 = \bar{\omega}$ is given by⁴

1. DISORDERED STATE (Unpolarized)



2. ORDERED STATE (Polarized)

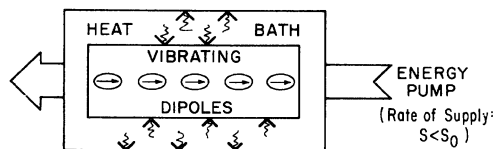


FIG. 1. Illustration of the Fröhlich model.

$$U(r) = -\frac{E}{r^3} + \frac{F}{r^6}, \quad (2)$$

where F is the London-van der Waals coefficient and E is the long-range constant given by⁴

$$E = \frac{\hbar e^2 Z |\bar{A}| \gamma}{4\bar{\omega} M} \left[\frac{1}{\epsilon'(\omega_+)} - \frac{1}{\epsilon'(\omega_-)} \right], \quad (3)$$

where M is the mass of a dipole, r is the lattice spacing, e is the electron charge, Z is the number of elementary charges on each dipole, and $\bar{A}$ is an angle constant. The quantities $\omega_{\pm}$ are the new dipole vibration frequencies which arise in the system as a result of the interactions

$$\omega_{\pm} = \left[\frac{1}{2}(\omega_1^2 + \omega_2^2) \pm \left[\frac{1}{4}(\omega_1^2 - \omega_2^2) + \frac{\beta_0^4}{(\epsilon'_{\pm})^2} \right]^{1/2} \right]^{1/2}, \quad (4)$$

where $\beta_0 = \gamma^2 e^2 \sqrt{Z_1 Z_2} / M r^3$ and γ is a coupling constant. Paul *et al.*^{12,13} presented a detailed derivation of the dielectric constant ϵ using the Wu-Austin Hamiltonian, Eq. (1), and obtained

$$\epsilon'(\omega) = 1 + \frac{4\pi^2}{kT} \rho d^2 f(\omega_{ij} n_i) |\chi|^2, \quad (5)$$

where ρ is the density of the substance, d is the average dipole moment, and f is a complicated function of the eigenfrequencies ω_i and occupation numbers n_i . Chatterjee *et al.*¹⁴ applied this result to cholesterol and found $\chi \sim 5 \times 10^{-3}$.

(ii) Bose condensation of the vibrational modes ω_i to one with the lowest allowed frequency ω_0 . This can be viewed as an energy redistribution in a coherent state with $\omega = \omega_0$, which is a polarized, metastable state and, consequently, has a nonzero dielectric polarization: $\langle \vec{P} \rangle \neq \vec{0}$. Further, Paul¹⁵ showed that the nonzero polarization is a manifestation of symmetry breaking associated with Bose condensations. He also demonstrated using Eq. (1) that $\langle a_l \rangle \neq 0$ and $\langle a_l^\dagger \rangle \neq 0$, where $\langle \rangle$ is the average employing the Hamiltonian H of Eq. (1). Since $\vec{d}_l \propto x_l \propto (a_l^\dagger + a_l)$, $\langle \vec{d}_l \rangle \neq \vec{0}$, where $\vec{d}_l$ is the dipole moment connected with the l th mode.

Experimental evidence in support of Fröhlich's conjecture about the existence of nonthermal excitations was found in Raman scattering studies of *Escherichia coli*^{16,17} and *Chlorella pyrenoidosa*.¹⁸ An indirect effect of increased yeast growth upon irradiation by specific frequencies of millimeter waves^{19,20} seems to indicate the importance of the Fröhlich-type excitations in growth control. Moreover, the interactions of human erythrocytes leading to the effect of rouleau formation can be understood using the Fröhlich theory.²¹⁻²⁶ In fact, Paul *et al.*¹² examined this phenomenon in detail and found good consistency with the theoretical model.

One question that has not been addressed in the studies involving the Fröhlich model is the propagation of order (polarization) throughout the system. However, a seemingly unrelated theory of quantum interactions in biological systems has been proposed by Davydov specifically to deal with this problem. We shall review the Davydov model in Sec. II.

II. THE DAVYDOV MODEL

Davydov and his associates, in a series of papers²⁷⁻³² and a book,³³ presented a model to explain the almost lossless energy transfer in quasi-one-dimensional biological substances such as protein and peptide chains and deoxyribonucleic acid (DNA). As an example, consider a peptide chain formed by H-N-C=O groups. Each peptide group has a constant electric dipole moment $d \sim 0.4$ debye. The neighboring groups interact through a dipole-dipole potential, Eq. (2), whose coefficients differ from those given by Fröhlich since the dipoles involved have no freedom of oscillations. The Davydov constant of attraction is

$$E = (3\cos^2\theta - 1)d^2, \quad (6)$$

where θ is the angle between the dipole $\vec{d}$ and the chain axis. The constant E reaches its maximum $2d^2$ when $\vec{d}$ is parallel to the axis. The excitation energy required for an electron to overcome the potential barrier is ~ 0.4 eV and electron transfer is facilitated by the presence of hydrogen bonds between the neighboring peptide groups. Furthermore, Davydov assumed strong coupling between the internal C=O vibrations and the elastic deformations of the chain.

The total Hamiltonian is written in the Heitler-London approximation as^{27,28}

$$H = \frac{1}{2} \sum_n (M \dot{x}_n^2 + \omega \rho_n^2) + \sum_n \left[(\epsilon - D_n) B_n^\dagger B_n - U(B_{n+1}^\dagger B_n + B_{n+1} B_n^\dagger) \right], \quad (7)$$

where

$$\rho_n = r - |x_n - x_{n-1}|, \quad (8)$$

$$D_n = D \left[1 + \frac{\beta}{r} \rho_n \right], \quad (9)$$

and D, β are positive parameters of the theory, and x_n is the x coordinate of the n th site of the chain. The electronic excitation operators B_n and $B_n^\dagger$ obey the Bose-

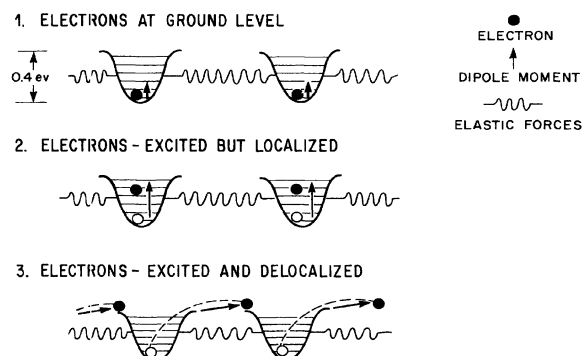


FIG. 2. Illustration of the Davydov model.

Einstein commutation relations.

Next, Davydov derived a nonlinear Schrödinger equation for the amplitude of the collective, electronic excitation $a(t)$ by using the Hamiltonian given above and the corresponding time-dependent Schrödinger equation for the wave function $|\phi(t)\rangle$:

$$i\dot{a}(t) - \lambda a(t) + U\nabla^2 a(t) + G_v |a(t)|^2 a(t) = 0. \quad (10)$$

Here, $|\phi(t)\rangle = \sum_n a_n(t) B_n^\dagger |0\rangle$ is a wave function of the total system and the remaining coefficients are

$$\lambda = \epsilon + T + U_H - D - 2U$$

and

$$G_v = \frac{\beta^2 D^2}{\omega r^2 [1 - (\nu/V)^2]}$$

with U_H representing the potential energy and T the kinetic energy of the lattice vibrations. The solutions of Eq. (10) are solitary waves of permanent profile which propagate along the chain with constant velocity ν (ν is less than the velocity of sound V).

Davydov's model was further elaborated by Scott³⁴ and applied to study the properties of the green algae: *Chlorella pyrenoidosa*.³⁵ The contraction mechanism of transversely strained animal muscles³⁶ was also elucidated with the use of this model. Moreover, it has recently been suggested³⁷ that this is applicable to the internal pulse in blood vessels.

We shall next explore the relationship between the Fröhlich and Davydov models. In order to compare the respective Hamiltonians [Eqs. (1) and (7)], certain transformations will be made to represent them in a compatible fashion. As a first step, we write the Davydov Hamiltonian using normal coordinates.

III. THE TRANSFORMATION OF DAVYDOV'S HAMILTONIAN TO NORMAL COORDINATES

The most apparent difference between the Wu-Austin and Davydov Hamiltonians is that the former employs normal coordinates whereas the latter is written in position coordinates. The transition from one set of coordinates to the other reflects the change from a localized to a collective picture of the excitations in the system. Davydov's Hamiltonian is transformed in the following steps.

(i) The potential energy of elastic vibrations is rewritten as

$$U_H = \frac{N\omega r^2}{2} - \omega r \sum_{n=1}^N |x_n - x_{n-1}| + \omega \sum_{n=1}^N x_n^2 - \omega \sum_{n=1}^N x_n x_{n-1}. \quad (11)$$

Since this represents a harmonic approximation, the length of the chain L is constant:

$$L = \sum_{n=1}^N |x_n - x_{n-1}|. \quad (12)$$

Using periodic boundary conditions the total energy of

elastic vibrations may be written as

$$H_{el} = \sum_{n=1}^N \left[\frac{p_n^2}{2M} + \sum_{n'=1}^N V_{nn'} x_n x_{n'} \right] + V_0, \quad (13)$$

where p_n denotes the momentum operator at the n th site and

$$V_{nn'} = \begin{cases} 2w & \text{if } n' = n \\ -w & \text{if } n' = n \pm 1 \\ 0 & \text{otherwise} \end{cases}. \quad (14)$$

The constant energy shift V_0 is

$$V_0 = NrW \left| \frac{r}{2} - \frac{L}{N} \right|. \quad (15)$$

Following a standard procedure,³⁸ the variables x_n and p_n are Fourier transformed to obtain

$$H_{el} = \sum_q \left[\frac{1}{2M} p_q p_q^\dagger + \frac{1}{2} M \Omega_q^2 x_q x_q^\dagger \right] + V_0. \quad (16)$$

Here, x_q and p_q are the Fourier transforms of x_n and p_n , respectively, and $\Omega_q = 2(W/M)^{1/2} \sin(q/2)$. Finally, defining the phonon annihilation operator as

$$b_q \equiv (2M\Omega_q)^{-1/2} (M\Omega_q x_q + ip_q^\dagger) \quad (17)$$

and the creation operator $b_q^\dagger$ as the Hermitian conjugate of b_q , a fully diagonalized, second-quantized operator for H_{el} is obtained:

$$H_{el} = \sum_q \Omega_q (b_q^\dagger b_q + \frac{1}{2}) + V_0. \quad (18)$$

(ii) The diagonal part of the electron excitation energy

$$H_{ex} = \sum_n [\epsilon - D(1 + \beta)] B_n^\dagger B_n \quad (19)$$

can be transformed into normal coordinates by introducing the operators a_q and $a_q^\dagger$ as the Fourier transforms of the operators B_n and $B_n^\dagger$, respectively:

$$H_{ex} = \sum_q \bar{\omega}_q a_q^\dagger a_q, \quad (20)$$

where $\bar{\omega}_q = \epsilon - D(1 + \beta)$. Note that $\bar{\omega}_q$ is independent of q .

(iii) The off-diagonal part of the electronic excitation energy

$$H_{off} = - \sum_n U (B_{n+1}^\dagger B_n + B_{n+1} B_n^\dagger) \quad (21)$$

is transformed in the same manner as H_{ex} into

$$H_{off} = - \sum_q U (2 \cos q a_q^\dagger a_q + 1). \quad (22)$$

Now, H_{ex} and H_{off} are combined to form the total electronic contribution to the Hamiltonian:

$$H_{exc} = H_{ex} + H_{off} = \sum_q (\omega_q a_q^\dagger a_q - U), \quad (23)$$

where the new energy eigenvalues are $\omega_q = \epsilon - D(1 + \beta) - 2U \cos q$. Note that ω_q is an even

function of q and $(\partial\omega_q/\partial q)_{q=0}=0$.

(iv) The remaining term of Eq. (7) represents the interactions between lattice vibrations and electronic excitations:

$$H_{\text{int}} = \sum_n D \frac{\beta}{r} |x_n - x_{n-1}| B_n^\dagger B_n, \quad (24)$$

and, with the use of the Fourier transformations introduced before,

$$H_{\text{int}} = \sum_{p,q,r} \chi_{pqr} (b_{-p}^\dagger + b_p) a_q^\dagger a_r, \quad (25)$$

where the coupling constant is

$$\chi_{pqr} = \frac{2D\beta}{r} \left[\frac{w}{N^2 M^3} \right]^{1/4} \delta(q,r) [\sin(q/2)]^{3/2}.$$

Finally, we write the entire Davydov Hamiltonian, Eq. (7), in normal coordinates as

$$H = H_0 + \sum_q \Omega_q b_q^\dagger b_q + \sum_q \omega_q a_q^\dagger a_q + \sum_{p,q,r} \chi_{pqr} (b_{-p}^\dagger + b_p) a_q^\dagger a_r, \quad (26)$$

where

$$H_0 = N \left[r w \left| \frac{r}{2} - \frac{L}{w} \right| - U \right] + \frac{1}{2} \sum_q \Omega_q, \quad (27)$$

$$\Omega_q = 2\sqrt{w}/M \sin(q/2), \quad (28)$$

$$\omega_q = \epsilon - D(1+\beta) - 2U \cos q, \quad (29)$$

$$\chi_{pqr} = \frac{2D\beta}{r} \left[\frac{w}{N^2 M^3} \right]^{1/4} [\sin(q/2)]^{3/2} \delta(q,r). \quad (30)$$

In Sec. IV we transform the Wu-Austin Hamiltonian, Eq. (1), to a form which is essentially identical to Eq. (26). As a result, we will be able to connect the two models by a single Hamiltonian.

IV. THE CANONICAL TRANSFORMATION OF THE WU-AUSTIN HAMILTONIAN

By comparing the normal coordinate representation of the Davydov Hamiltonian, Eq. (26), with the Wu-Austin Hamiltonian, Eq. (1), many similarities may be seen. However, the main differences that the Wu-Austin Hamiltonian involves are the interactions of the cell phonons with the energy pump (i.e., the terms proportional to $P_i a_j^\dagger$ and $P_i^\dagger a_j$) as well as first-order interactions with the heat

bath (i.e., the terms proportional to $b_i a_j^\dagger$ and $b_i^\dagger a_j$). Fortunately, this problem may be eliminated by a canonical transformation of the Wu-Austin Hamiltonian and obtaining an equivalent operator in a form identical to the Davydov Hamiltonian.

Assuming that the *second-order* cell-phonon-heat-bath interactions are sufficiently small perturbations, we neglect them for the moment and rewrite Eq. (1) as

$$H = H_0 + H_I, \quad (31)$$

where H_0 is the unperturbed part of the Hamiltonian

$$H_0 = \sum_i \omega_i a_i^\dagger a_i + \sum_i \Omega_i b_i^\dagger b_i + \sum_i \theta_i p_i^\dagger p_i \quad (32)$$

and H_I contains the *first-order* perturbation terms:

$$H_I = \sum_{i,j} (\lambda b_i a_j^\dagger + \lambda^* b_i^\dagger a_j) + \sum_{i,j} (\xi P_i a_j^\dagger + \xi^* P_i^\dagger a_j). \quad (33)$$

In order to effectively *decouple* the cell operators $a_i^\dagger, a_i$ from the heat-bath operators $b_i^\dagger, b_i$ and the energy-pump operators $P_i^\dagger, P_i$, we use the following unitary transformation:

$$\tilde{H} = e^{-S} H e^S \quad (34)$$

with the matrix S defined as

$$S = \sum_{i,j} (\mu b_i^\dagger a_j + \nu b_i a_j^\dagger) + \sum_{i,j} (\Delta P_i a_j^\dagger + \sigma P_i^\dagger a_j). \quad (35)$$

Applying the Baker-Hausdorff formula

$$e^{-S} H e^S = H_0 + H_I + [H_0, S] + [H_I, S] + \frac{1}{2} [[H_0, S], S] + \frac{1}{2} [[H_I, S], S] + \dots \quad (36)$$

together with the requirement

$$H_I = -[H_0, S] \quad (37)$$

yields the following values of the transformation coefficients:

$$\mu = \frac{\lambda}{\Omega_i - \omega_j}, \quad \nu = \frac{\lambda^*}{\omega_j - \Omega_i}, \quad (38)$$

$$\Delta = \frac{\xi}{\theta_i - \omega_j}, \quad \sigma = \frac{\xi^*}{\omega_j - \theta_i}. \quad (39)$$

Therefore, $\sigma = -\Delta^*$ and $\nu = -\mu^*$. Consequently, the series expansion, Eq. (36), is evaluated up to the fifth term as

$$\tilde{H} = H_0 + \Delta H, \quad (40)$$

where ΔH is the term representing the net effect of the first-order perturbations

$$\begin{aligned} \Delta H = & \frac{1}{2} \sum_{i,j} \left\{ \sum_k \left[|\xi|^2 \left(\frac{1}{\omega_i - \theta_k} + \frac{1}{\omega_j - \theta_k} \right) + |\lambda|^2 \left(\frac{1}{\omega_i - \Omega_k} + \frac{1}{\omega_k - \Omega_k} \right) \right] \right\} a_i^\dagger a_j \\ & + \frac{1}{2} \sum_{i,j} \left\{ \sum_k \left[|\xi|^2 \left(\frac{1}{\theta_i - \omega_k} + \frac{1}{\theta_j - \omega_k} \right) \right] \right\} P_i^\dagger P_j + \frac{1}{2} \sum_{i,j} \left\{ \sum_k \left[|\xi|^2 \left(\frac{1}{\Omega_j - \omega_k} + \frac{1}{\Omega_j - \omega_k} \right) \right] \right\} b_i^\dagger b_j. \quad (41) \end{aligned}$$

Although Eq. (41) contains off-diagonal terms, ΔH remains Hermitian. Moreover, ΔH preserves the total number of quasiparticles in the system. The sixth term in the expansion, Eq. (36), reintroduces the coupling between the cell phonons and the heat bath and the energy pump. However, the new coupling coefficients are proportional to $|\xi|^2\xi$ and $|\lambda|^2\lambda$ this time. Repeating the canonical transformation procedure removes these couplings beyond the fifth term in the new expansion according to Eq. (36). consequently, the operator ΔH after the second transformation still preserves the initial form given by Eq. (41) save for the energy constants which are supplemented by corrections proportional to $|\xi|^4$ and $|\lambda|^4$. Carrying out this procedure up to infinity brings the following form of $\tilde{H}$:

$$\tilde{H} = \sum_{i,j} \omega_{ij} a_i^\dagger a_j + \sum_{i,j} \Omega_{ij} b_i^\dagger b_j + \sum_{i,j} \theta_{ij} P_i^\dagger P_j, \quad (42)$$

where

$$\begin{aligned} \omega_{ij} = & \omega_i \delta_{ij} + \frac{1}{2} \sum_k \left[|\xi|^2 \left(\frac{1}{\omega_i - \theta_k} + \frac{1}{\omega_j - \theta_k} \right) \right. \\ & \left. + |\lambda|^2 \left(\frac{1}{\omega_i - \Omega_k} + \frac{1}{\omega_j - \Omega_k} \right) \right] \\ & + \sum_{n \geq 2} (|\xi|^{2n} + |\lambda|^{2n}), \end{aligned} \quad (43)$$

where $\sum_{n \geq 2} (|\xi|^{2n})$ represents a series in $|\xi|^{2n}$, with $n=2,3,\dots,\infty$ and $\sum_{n \geq 2} (|\lambda|^{2n})$ a series in $|\lambda|^{2n}$, with $n=2,3,\dots,\infty$;

$$\begin{aligned} \Omega_{ij} = & \Omega_i \delta_{ij} + \frac{1}{2} \sum_k |\lambda|^2 \left[\frac{1}{\Omega_i - \omega_k} + \frac{1}{\Omega_j - \omega_k} \right] \\ & + \sum_{n \geq 2} (|\lambda|^{2n}); \end{aligned} \quad (44)$$

$$\begin{aligned} \theta_{ij} = & \theta_i \delta_{ij} + \frac{1}{2} \sum_k |\xi|^2 \left[\frac{1}{\theta_i - \omega_k} + \frac{1}{\theta_j - \omega_k} \right] \\ & + \sum_{n \geq 2} (|\xi|^{2n}). \end{aligned} \quad (45)$$

The applicability of this procedure rests entirely on the detailed energy structure and the strength of the coupling constants which determine the radii of convergence of the series involved in Eqs. (43)–(45) denoted: ξ_1, λ_1 and ξ_2 and λ_2 , respectively. All the energy constants converge if $|\xi| < \min(\xi_1, \xi_2)$ and if $|\lambda| < \min(\lambda_1, \lambda_2)$. Since the heat and energy reservoirs are assumed much larger than the cell, we approximate $\Omega_{ij} \simeq \Omega_i \delta_{ij}$ and $\theta_{ij} \simeq \theta_i \delta_{ij}$. Furthermore, in the process of quantum averaging the off-diagonal terms (those with $i \neq j$) will disappear and the entire Wu-Austin Hamiltonian may be finally presented as

$$\begin{aligned} H = & H'_0 + \sum_i \omega_i a_i^\dagger a_i + \sum_i \Omega_i b_i^\dagger b_i \\ & + \frac{1}{2} \sum_{i,j,k} (\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k), \end{aligned} \quad (46)$$

where $H'_0 = H_0 + \sum_i \theta_i P_i^\dagger P_i$ since the energy-pump operator commutes with all the remaining operators in Eq. (46). The new cell energy constant is $\bar{\omega}_i = \omega_i \delta_{ij}$ where ω_{ij} is that of Eq. (43). This Hamiltonian, Eq. (46), contains the second-order cell-phonon–heat-bath interactions as the only coupling terms. The effect of all first-order interactions is incorporated into the new energy constant $\bar{\omega}_i$.

It is concluded that, provided the interactions between the subsystems are sufficiently small, the Wu-Austin Hamiltonian can be successfully and accurately transformed into an effective operator which is exactly of the same form as the Davydov Hamiltonian, Eq. (26). Therefore, Eq. (46) can be considered a prototype Hamiltonian embracing both theories. Before deriving equations of motion for the operators involved in this Hamiltonian, another canonical transformation will be applied in order to clarify further the role of the remaining interaction term.

V. THE CANONICAL TRANSFORMATION OF THE MODEL HAMILTONIAN

We have seen thus far that the two seemingly unrelated theories of biological order can, indeed, be described using the same Hamiltonian

$$H = H_0 + H_I, \quad (47)$$

where

$$H_0 = H'_0 + \sum_i \bar{\omega}_i a_i^\dagger a_i + \sum_i \Omega_i b_i^\dagger b_i \quad (48)$$

and

$$H_I = \frac{1}{2} \sum_{i,j,k} (\chi b_{-k}^\dagger + \chi^* b_{+k}) a_i^\dagger a_j. \quad (49)$$

We have assumed that the indices i, j , and k refer to conserved quantities. Equation (47) can be regarded as a model Hamiltonian for the phenomena described by both theories. It employs two types of creation and annihilation operators: $\{a_i^\dagger, a_i\}$ and $\{b_i^\dagger, b_i\}$. The first type of operators refers to the active modes of the cell: dipole vibrations in the Fröhlich model or electronic excitations in the Davydov model. The second type of operators belongs to those of the thermal environment as displayed by the heat bath in the Fröhlich model or the elastic lattice vibrations in the Davydov model. Both types of operators are coupled through a second-order interaction term, Eq. (49), and it is assumed that the heat reservoir has a much larger energetic capacity than the cell. Consequently, only the cell will be noticeably affected by the coupling. The particular (dispersion) properties of the energy constants ω_i and Ω_i and of the coupling constant χ remain a characteristic feature of each model. In general, χ is assumed to be a function of i and j since the third index k is determined from a conservation law: $Q(i) - Q(j) - q(k) = 0$, where Q and q refer to a conserved quantity (e.g., momentum) of the cell and heat-bath quasiparticle, respectively.

In order to see the role of the interaction term, Eq. (49), better, the model Hamiltonian, Eq. (47), is replaced by its

equivalent, in much the same way as was applied to the Wu-Austin Hamiltonian in Sec. IV. Choosing the transformation matrix S as

$$S = \sum_{i,j,k} (Ab_{-k}^\dagger + Bb_{+k})a_i^\dagger a_j \quad (50)$$

and using a Baker-Hausdorff expansion, Eq. (36), the transformation coefficients A and B are found as

$$A = -\frac{\chi_{ij}}{(\omega_i - \omega_j) + \Omega_{-k}}, \quad B = -\frac{\chi_{ij}^*}{(\omega_i - \omega_j) - \Omega_k}. \quad (51)$$

The first nonvanishing correction term in the series, Eq. (36), is

$$\Delta H_I = \sum_{i,j,k} \Delta_{ijk} a_i^\dagger a_j a_k^\dagger a_{i-j+k}, \quad (52)$$

where

$$\Delta_{ijk} = |\chi_i|^2 \frac{\Omega_{j-k}}{(\omega_j - \omega_k)^2 - \Omega_{j-k}^2}.$$

This is referred to as the Hartree-Fock (or the mean-field) approximation in which the effect of the external environment is averaged out to form a homogeneous background. The coupling between both modes of excitation is reintroduced into the Hamiltonian in the sixth-order term of the expansion, Eq. (36). The same procedure may be reapplied to remove this coupling and, as a result, correction terms involving an increasing number of second-quantized operators will be added to Eq. (52). However, on the assumption that the coupling constant χ is, relatively, of small magnitude, the higher-order terms are neglected as they are proportional to $|\chi|^{2n}$ with $n=2,3,\dots$

We can now separate the model Hamiltonian H into two commuting (noninteracting) parts H' and H'' ,

$$H = H' + H'', \quad (53)$$

where H' is an equivalent operator representing the combined effect of the surroundings on the active modes of the cell. Using the more conventional wave number indices,

$$H' = \sum_k W_k a_k^\dagger a_k + \sum_{k,k',q} \Delta_{kk'q} a_{k'+q}^\dagger a_{k-q}^\dagger a_k a_{k'}, \quad (54)$$

where

$$W_k = \bar{\omega}_k - \left[\frac{|\chi_0|^2}{\Omega_0} N \right], \quad (55)$$

N is the total number of cell phonons, and $\Delta_{kk'q}$ is the effective coupling constant,

$$\Delta_{kk'q} = |\chi_{kk'}|^2 \frac{\Omega_q}{(\omega_k - \omega_{k'})^2 - (\Omega_q)^2}. \quad (56)$$

H'' describes the state of the external environment,

$$H'' = H'_0 + \sum_q \Omega_q b_q^\dagger b_q. \quad (57)$$

Since $[H', H''] = 0$, H'' can be henceforth neglected in the discussion. H' contains all the relevant information about the perturbation effects on the biosystem.

VI. THE EQUATION OF MOTION

Having derived the model Hamiltonian, Eq. (53), which combines both theories of biological order, we shall investigate the ensuing time-dependent phenomena. In general this is a very complicated problem. Fortunately, however, the Hamiltonian presents certain features that can be used to advantage. It is easy to see that, if in each of the creation and annihilation operators in Eq. (53) we make the following transformations:

$$a_k \rightarrow e^{i\theta_k} a_k \quad (58)$$

$$a_k^\dagger \rightarrow e^{-i\theta_k} a_k^\dagger, \quad (59)$$

(where θ_k is some arbitrary phase factor), the Hamiltonian H' does not remain invariant. It has been shown by Umezawa and co-workers³⁹ that such a lack of invariance signifies the phenomenon of symmetry breaking. Paul¹⁵ has shown that this phenomenon is accompanied by a field translation of the operators

$$a_k \rightarrow a_k + c_k, \quad a_k^\dagger \rightarrow a_k^\dagger + c_k^*.$$

Here, the quantities c_k and c_k^* are not operators but c -numbers. The occupation number states upon which the operators a_k and $a_k^\dagger$ act are transformed into the Glauber coherent states. Thus

$$|n_1, n_2, \dots\rangle \rightarrow |c_1, c_2, \dots\rangle \equiv |\{c\}\rangle.$$

The $|\{c\}\rangle$ are the Glauber coherent states and are eigenstates of the annihilation operators

$$a_k |\{c\}\rangle = c_k |\{c\}\rangle. \quad (60)$$

The adjoint of this equation yields

$$\langle \{c\} | a_k^\dagger = c_k^* \langle \{c\} |. \quad (61)$$

The Hamiltonian for the system within the coherent-state description is given by the expectation of H' [as given by Eq. (54)]. We will adopt the symbol $\mathcal{H}'$ to designate this expectation value:

$$\mathcal{H}' = \frac{\langle \{c\} | H' | \{c\} \rangle}{\langle \{c\} | \{c\} \rangle} \equiv \mathcal{H}'(\{c\}, \{c^*\}). \quad (62)$$

This can be readily calculated by substituting Eq. (54) in Eq. (62) and employing Eqs. (60) and (61). We now have

$$\mathcal{H}'(\{c\}, \{c^*\}) = \sum_k W_k c_k^* c_k + \sum_{k,k',q} \Delta_{kk'q} c_{k'+q}^* c_{k-q}^* c_k c_{k'}. \quad (63)$$

In this representation the Hamiltonian is no longer an operator but a c number depending upon the field-translation parameters. The equations of motion for the quantities $\{c\}$ and $\{c^*\}$ can be obtained from Hamilton's equations

$$i\dot{c}_l = \frac{\partial \mathcal{H}'}{\partial c_l^*}, \quad -i\dot{c}_l^* = \frac{\partial \mathcal{H}'}{\partial c_l}. \quad (64)$$

By inserting $\mathcal{H}'$ from Eq. (62) and carrying out the differentiation we get, along with its complex conjugate,

$$i\dot{c}_l = W_l c_l + \sum_{k,k'} Y_{kk'} c_{k+k'}^\dagger c_k c_{k'}, \quad (65)$$

where

$$Y_{kk'} = \Delta_{k,k',l-k'} + \Delta_{k,k',k-l} . \quad (66)$$

Similarly, an equation of motion for c_l^* can be derived.

It is easy to see that within the coherent-state description the expectation value of the annihilation and creation operators do not vanish, but propagate throughout the

system without loss of energy. Before demonstrating these solutions (in Sec. III) it should be mentioned that if only one of the following conditions is met, ordinary waves exhibiting diffusion will result instead of solitons.

$$\frac{\langle \{c\} | a_k | \{c\} \rangle}{\langle \{c\} | \{c\} \rangle} = c_k^* , \quad (67)$$

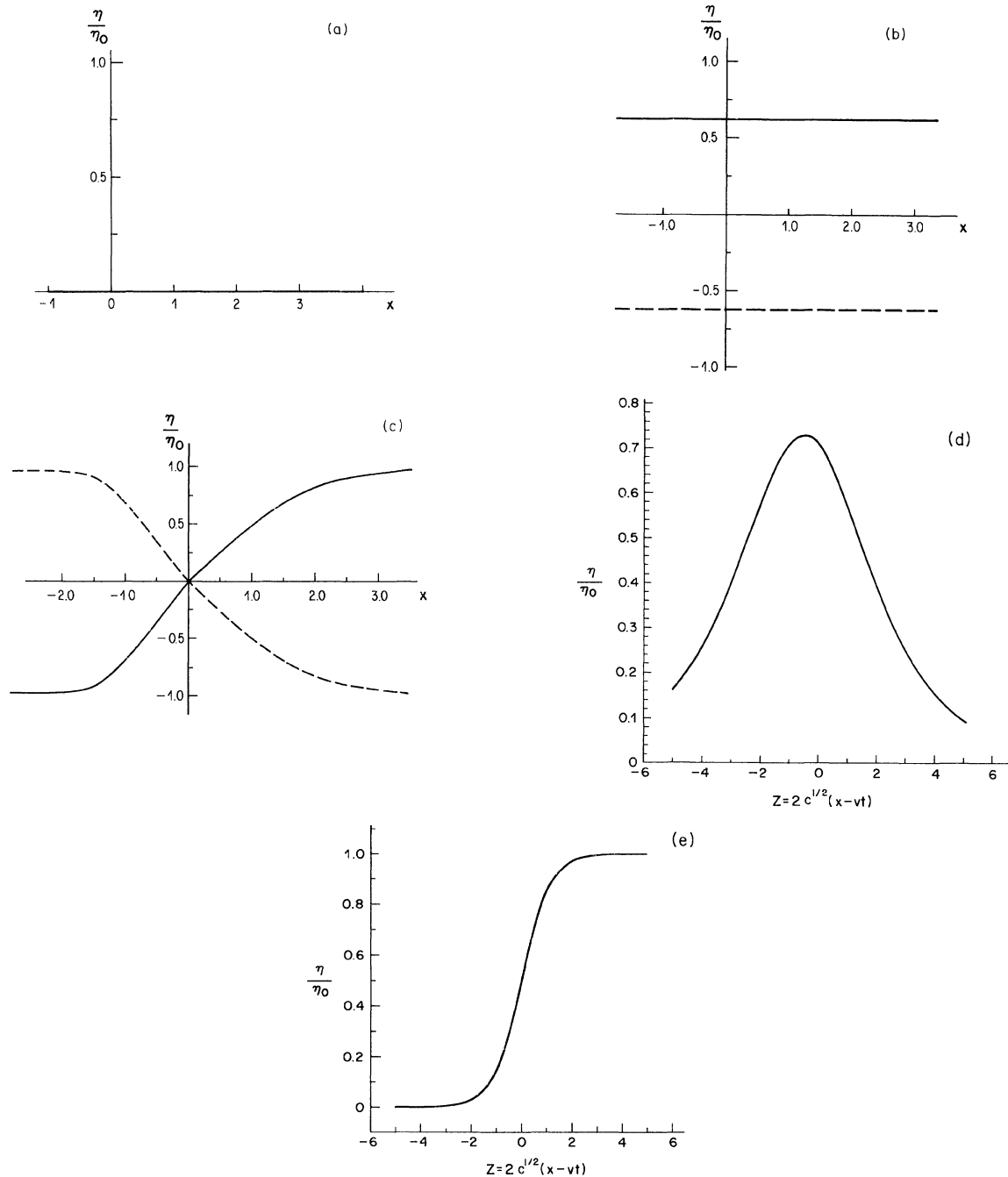


FIG. 3. (a) Solution 1 (static) representing the disordered phase. (b) Solution 2 (static) representing the two possible ordered phases. (c) Solution 3 (static) representing domain walls between the two possible ordered phases. (d) Solution 4 (dynamic) representing the kink-antikink couple. (e) Solution 5 (dynamic) representing the moving kink connecting the ordered and disordered phases.

(i) The third- or a higher-order term in the expansion of W_k in a series of k is significant and cannot be neglected.

(ii) The first-order term W_1 does not vanish, as, for example, in the case of acoustic phonons where $W(k) \propto \sin(k/2)$.

(iii) The Bose condensation does not occur and the averages over $|0\rangle$, and not $|0\rangle\rangle$, must be taken. Then a linear differential equation in the form of a diffusion equation will be obtained:

$$i\dot{\eta} = W_0\eta + W_2\eta_{xx}. \quad (68)$$

The amplitudes of the solutions to (68) decay exponentially with x in direct proportionality to the diffusion constant W_2 .

VII. SOLUTIONS

The present equation, Eq. (65), is a special case of a nonlinear Schrödinger equation and is solved following the detailed derivation⁴⁰ for a general case. The parameters in Eq. (65) have a physical interpretation which depends on the particular model used. In both cases, however, the nonlinearity parameter Y is proportional to the strength of the cell-heat reservoir coupling: $|\chi|^2$. In the Davydov model the diffusion constant W_2 equals the dipole-dipole potential U and the linear coefficient $W_0 = \epsilon - D(1+B) - 2U - (|\chi_0|^2/\Omega)N$.

All solutions of Eq. (65) can be divided into two groups: (i) static (or time-independent) and (ii) dynamic (or time-dependent). The static solutions follow.

(1) $\eta_1(x) \equiv 0$. It exists when $W_0 \geq 0$ and the cell-heat-bath coupling increases the value of W_0 since Ω_k are negative. This solution represents a disordered, unpolarized phase since the field translation is proportional to the net

polarization. When $\eta_1(x) = 0$, there is neither Bose condensation nor a net dielectric polarization.

(2) A solution corresponding to the homogeneous, ordered phase whose amplitude is

$$\eta_2(x, t) = \pm [(-W_0)/T]^{1/2} \quad (69)$$

and it can only exist when $W_0 \leq 0$ which is facilitated by the presence of a sufficiently strong coupling:

$$|\chi|^2 \geq \frac{\omega_0}{N} \Omega_0.$$

This solution represents an ordered, polarized phase existing when Bose condensation takes place. Solutions (1) and (2) were, in fact, the subject of investigations by Fröhlich and they describe the time-independent aspect of the problem. Note that we use the terminology of phase transitions since the field translation η can be treated as a time- and space-dependent order parameter. Moreover, since without the presence of external electric fields breaking the symmetry manifested by the appearance of “ $\pm$ ” in Eq. (69) two ordered phases are allowed, there is an additional solution connecting the two phases. It has the form of a domain wall and is given by

(3) an interface solution,

$$\eta_3(x, t) = \eta_0 \tanh \left[\frac{x}{\sqrt{2}\xi_c(T)} \right], \quad (70)$$

where $\eta_0 = |\eta_2|$ and $\xi_c(T)$ is the so-called correlation length which equals $\xi_c(T) = -W_2/W_0$. Thus, it is necessary that $W_2 > 0$ and $W_0 < 0$ for this solution to exist.

The dynamic solutions are solitons.

(4) A kink-antikink pair corresponding to the Davydov soliton:

$$\begin{aligned} \eta_4(x, t) = \exp[i\delta_c(\Delta x_c - v_c t)] \{ [(1+A)/4C] \sinh[x_1 \pm 2\sqrt{C}(x - vt)] \\ + [(1-A)/4C] \cosh[x_1 \pm 2\sqrt{C}(x - vt) - D/2C] \}^{-1/2}, \end{aligned} \quad (71)$$

where δ_c , x_c , and v_c are the carrier width, phase shift, and velocity, respectively, and x_1 is the envelope phase shift. The following symbols have also been introduced:

$$C = \frac{-W_0}{W_2} + \frac{1}{2} \left[\frac{v}{W_2} \right]^2, \quad (72)$$

$$A = - \left[\frac{T}{W_2} \right]^2 \leq 0, \quad (73)$$

$$D = \frac{-T}{W_2}. \quad (74)$$

This solution can be interpreted as a solitary wave of polarization-soliton propagating through the system at a constant velocity v without loss of energy. Both directions of motion are allowed and the width of this soliton is $\delta_e = 2\sqrt{C}$ which vanishes as $W_0 \rightarrow 0$ or $W_2 \rightarrow \infty$.

(5) This solution is a moving order-disorder phase wall given by

$$\begin{aligned} \eta_5(x, t) = \exp[i\delta_c(\Delta x_c - v_c t)] \\ \times \{ (1/2C) \exp[x_2 \pm 2\sqrt{C}(x - vt)] - D/2C \}^{-1/2}. \end{aligned} \quad (75)$$

It exists only when $A = 0$ (i.e., if $T = 0$ or $W_2 = \infty$) and $C > 0$. This solution separates two neighboring regions, one of which is disordered (solution 1) and the other of which is ordered (solution 2) and therefore exists at the transition point where $W_c = 0$.

Neither the static domain wall (solution 3) nor the dynamic domain wall (solution 5) have been found before and they seem to play important roles in these phenomena. Moreover, the remaining solutions which are divided into two distinct groups appear to describe different aspects of the same phenomenon. The static solutions described first by Fröhlich are manifestations of the Bose condensation in the system and the dynamic solutions are polarization waves propagating across the system as soli-

tons in a lossless manner. Therefore, the two seemingly different models of biological order are not only mathematically equivalent but also physically complementary. We have shown that they can be unified into a general, complete theory.

VIII. CONCLUSIONS

Using a series of Fourier and canonical transformations, we have demonstrated the relationship between the two seemingly independent theories of biological order put forward by Fröhlich and Davydov. Within the bounds of the approximations used, the two models are mathematically equivalent as they yield identical effective Hamiltonians. A model Hamiltonian encompassing both models is constructed which yields an equation of motion for the ground-state expectation value of the creation and annihilation operators involved, called a field translation. Under the conditions for Bose condensation the Fourier transform of the field-translation variable is shown to obey a nonlinear Schrödinger equation (NLSE) whose solutions are well known. The set of NLSE solutions can be classified as static (time independent) or dynamic (time dependent). The first group was the subject of Fröhlich's studies whereas the second was discussed by Davydov. Thus, the two models seem to complement each other physically. We have found two new solutions: a standing interface between two ordered phases whose amplitudes are of opposite sign and a moving interface between an ordered and a disordered phase. It is concluded that the field translation associated with the presence of Bose condensation can be considered as an order parameter in biological systems. In the absence of Bose condensation or when the dispersion function for the cell phonons does not meet certain restrictions, the resultant equation of motion does not admit solitons, but ordinary waves which diffuse and decay.

Based on our findings, we expect both static and dynamic effects to occur simultaneously and they are interpreted as two aspects of the same phenomenon. Thus, Bose condensation of membrane-vibrational modes should be accompanied by the solitonic propagation of a polarization wave. Conversely, the soliton transport of bond energy along the peptide chain should be accompanied by Bose condensation of lattice vibrations. We note that the existence of solitons and Bose condensation have been

shown to be related^{41,42} also elsewhere, which supports our conclusions.

The main difference between the Davydov and Fröhlich models is one of emphasis on different manifestations of the same problem. Fröhlich's model concentrates on time-independent effects (stable states) and leads to conditions for establishing order in the system whereas Davydov's model investigates time-dependent effects and highlights propagation of this order throughout the system. Furthermore, due to the differences in the particulars of the systems under study, the energy constants and coupling coefficients appear to have different strengths and dispersion properties. Notably, Fröhlich assumed the dipoles to vibrate at high frequencies and his approach implies⁴³ that the dipole-dipole energy is less than kT . Davydov, on the other hand, considers the case when the dipoles are statically aligned and their interaction energy far exceeds kT .⁴⁴ Both models, however, arrive at an attractive long-range effective potential which is proportional to r^{-3} . Since the active objects are of different dimensions in these theories (Fröhlich, membranes; Davydov, electrons), it is not surprising to find that the two potentials differ significantly in strength. Paul *et al.*¹⁸ calculated, using the Wu-Austin Hamiltonian, that $E \simeq -2.3 \times 10^{-36} \text{ Jm}^3$ for human erythrocytes whereas for the peptide group $E \simeq -2.0 \times 10^{-60} \text{ Jm}^3$ using the Davydov expression. It would be necessary to investigate the dispersion properties of the coefficients appearing in the Wu-Austin Hamiltonian, without which it would be difficult to argue that solitons do exist in biological membranes.

We note in closing that Fröhlich⁴³ himself predicted the significance of electronic excitations in biological cells although he did not relate them explicitly to the model of membrane vibrations. Having related Fröhlich's theory to Davydov's theory of electron transport it seems natural to explore the possibilities of membranes and electrons attaining similar frequencies. Such an analysis could quantify Fröhlich's speculations about the role of free electrons in destroying the growth control mechanisms.⁴³

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